

Q1.

Application layer:

1. HTTP
2. DNS
3. SSDP
4. DHCP
5. DHCPv6
6. MDNS

7. LLNMR

Transport layer:

UDP
TCP

Network layer:

ARP
ICMPV6
IGMPV3

Q2.

a. Total data received when accessing `http://neverssl.com` = approximately 51MB.

b. Total data received when accessing `http://neverssl.cornell.edu` = 52328

Q3 (A) Number of HTTP GET requests observed: 4
Get Requests: 4

- GET /edgedl/diffgen-puffin/...
- GET /online HTTP/1.1
- GET /favicon.ico HTTP/1.1
- GET /edgedl/diffgen-puffin/...

(B) What version of HTTP is the server running?

The server is running HTTP/1.1.

C. (i) Total TCP segments received: 52328

iii) Bytes $A \rightarrow B = 1 \text{ MB}$ (client $\rightarrow$ server)
Bytes $B \rightarrow A = 51 \text{ MB}$ (server $\rightarrow$ client)

The total amount of data received in the corresponding HTTP response is approximately 51 MB.

3.(D) TCP Three way Handshake

Packet	Source $\rightarrow$ Destination	Sequence Number
SYN	10.85.19.25 $\rightarrow$ 34.104.35.123	Seq = 0
SYN-ACK	34.104.35.123 $\rightarrow$ 10.85.19.25	Seq = 0, ACK = 1
ACK	10.85.19.25 $\rightarrow$ 34.104.35.123	Seq = 1, ACK = 1